ADDRESSING TYPES (IP, MAC & Port)

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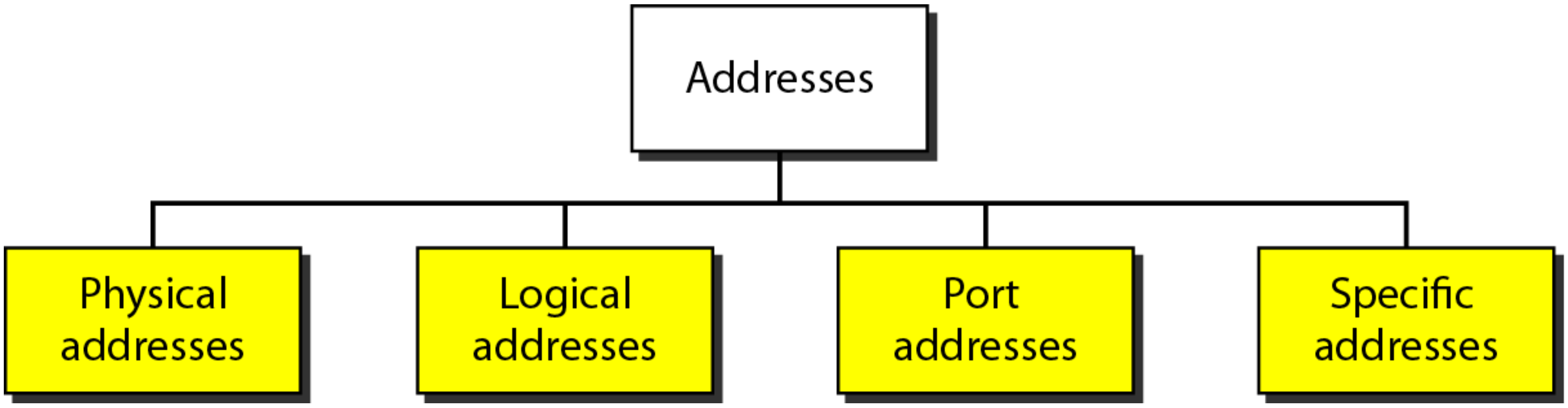
TSEC

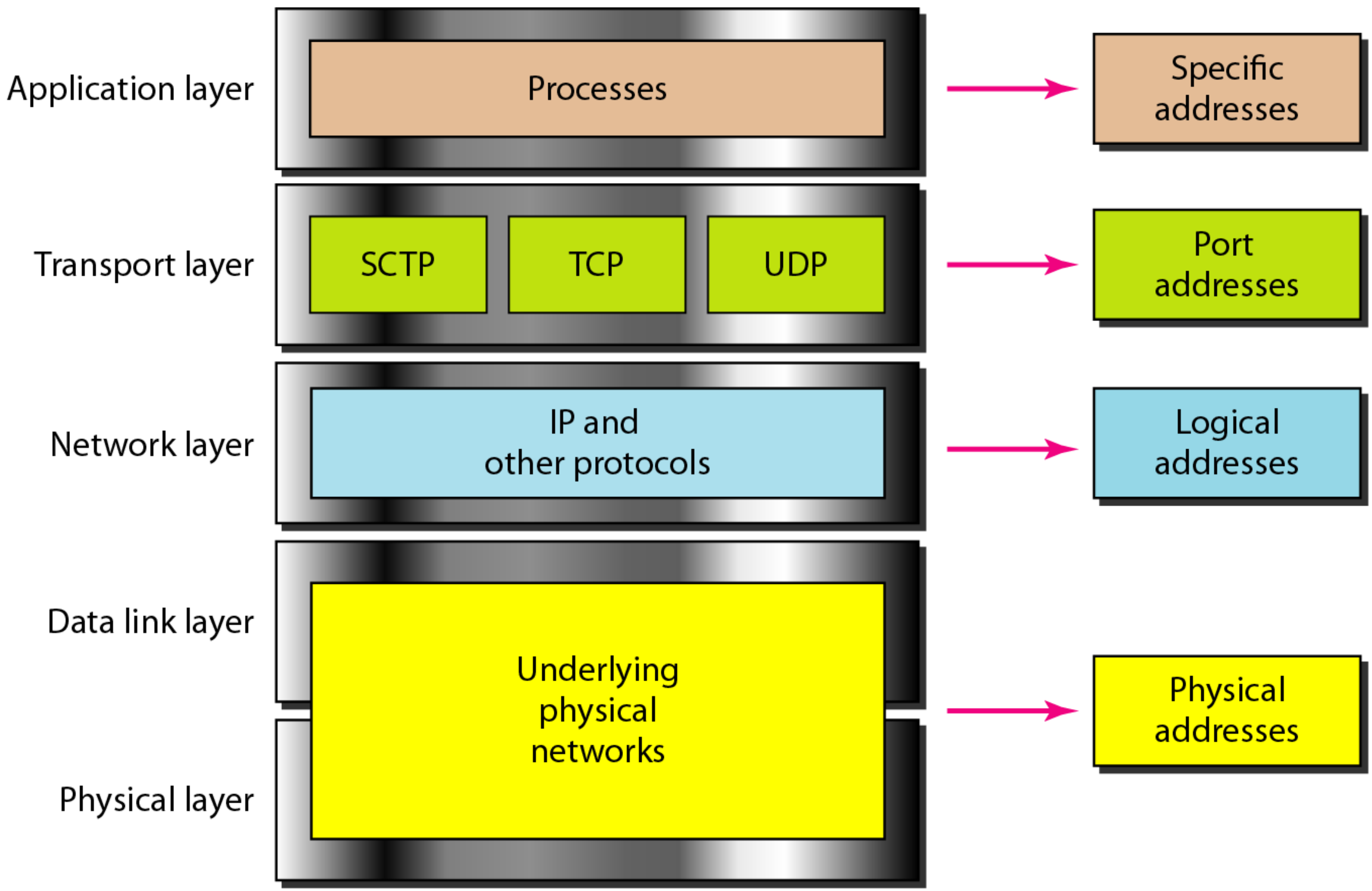
ADDRESSING



Four levels of addresses used in an internet employing the *TCP/IP* protocols:

- **Physical (link) addresses**
- **Logical (IP) addresses**
- **Port addresses &**
- **Specific addresses**

Figure 2.18 *Relationship of layers and addresses in TCP/IP*



ADDRESSING

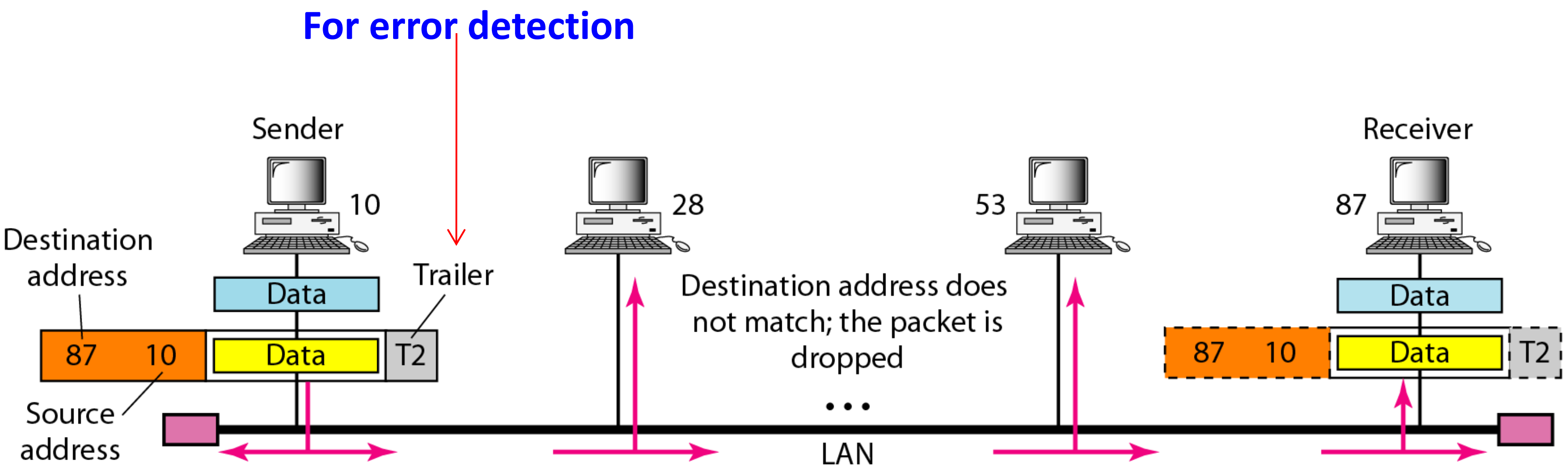
1. Physical (or Link)Addresses

- Address of a node as defined by its LAN or WAN
- Included in the frame used by the data link layer
- **Lowest-level** address
- **Size** and **format** vary depending on the network

Example:

- Ethernet uses a **6-byte (48-bit)** physical address that is imprinted on the **network interface card** (**NIC**)
- LocalTalk (Apple), has a **I-byte** dynamic address that changes each time the station comes up

Figure 2.19 *Physical addresses*



Note that in most data link protocols, the destination address (87 in this case) comes before the source address (10 in this case).

Example 2.1

- *In **Figure 2.19** a node with physical address 10 sends a frame to a node with physical address 87.*
- *The two nodes are connected by a link (bus topology LAN).*
- *The computer with physical address 10 is the sender, and the computer with physical address 87 is the receiver.*

Example 2.1

- *The data link layer at the sender receives data from an upper layer. It **encapsulates** the data in a frame, adding a **header** and a **trailer**.*
- *The header, among other pieces of information, carries the receiver and the sender physical addresses.*
- *Note that in most data link protocols, the destination address (87 in this case) comes before the source address (10 in this case).*
- *Shown is a bus topology for an isolated LAN. In a bus topology, the frame is propagated in both directions (left and right).*

Example 2.1

- *The frame propagated to the left **dies** when it reaches the end of the cable if the cable end is terminated appropriately.*
- *The frame propagated to the right is* sent to every station on the network. Each station with a physical addresses other than 87 drops the frame because the destination address in the frame does not match its own physical address.
- The intended destination computer, however, finds a match between the destination address in the frame and its own physical address.
- The frame is checked, the **header** and **trailer** are **dropped**, and the data part is **decapsulated** and delivered to the upper layer.

Example 2.2

*Most local-area networks use a **48-bit** (**6-byte**) physical address written as **12 hexadecimal digits**; every byte (2 hexadecimal digits) is separated by a **colon**, as shown below:*

07:01:02:01:2C:4B

A 6-byte (12 hexadecimal digits) physical address.

ADDRESSING

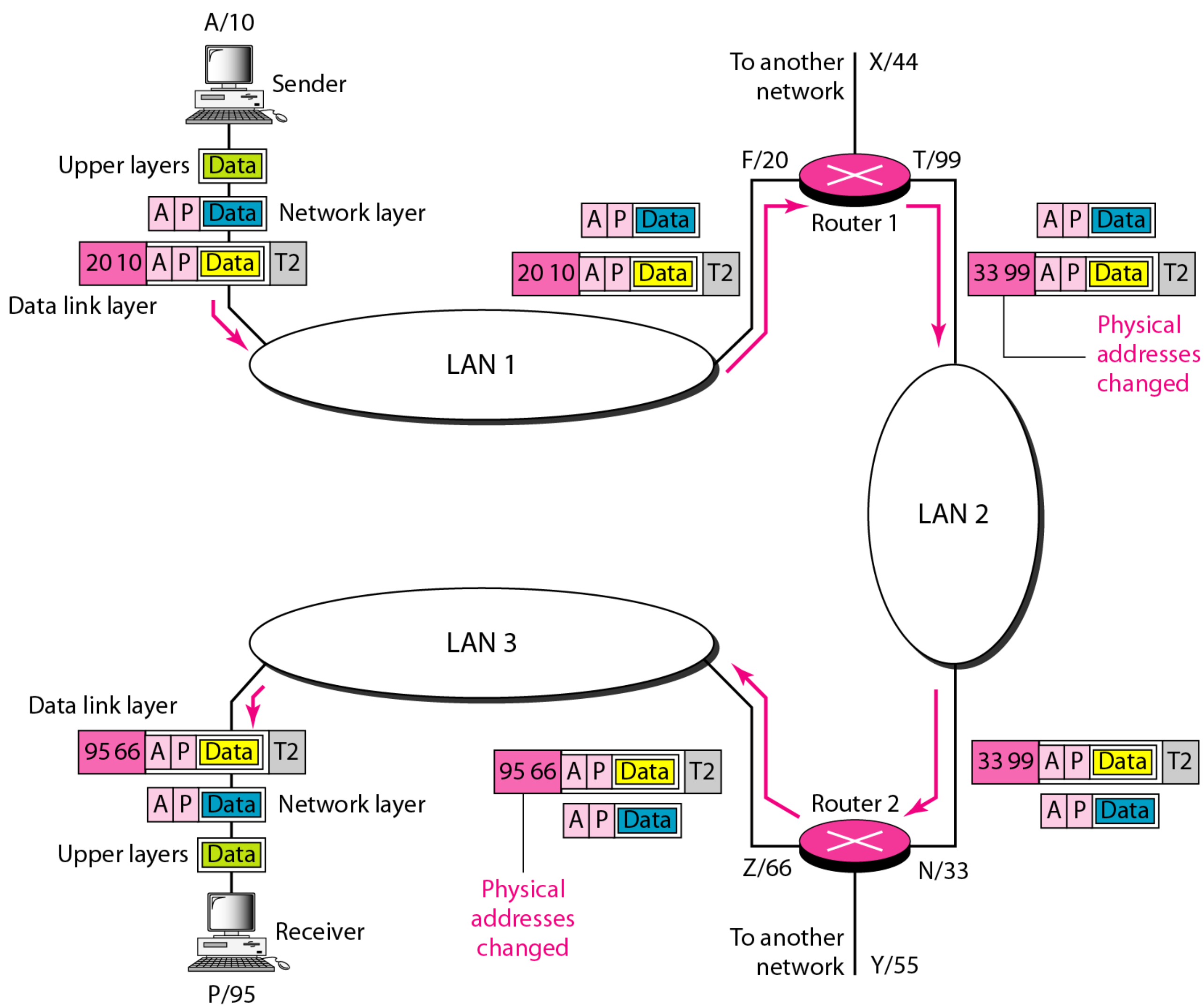
2. Logical Addresses

- Provide a **universal addressing system** in an internetwork environment
- **Allows the host to be identified** uniquely, **regardless of** the underlying **physical network**
- A logical address also called IP address in the Internet is currently a 32-bit address
- **No two** publicly addressed and visible hosts on the Internet can have the same IP address

Example 2.3

- ***Figure 2.20*** *shows a part of an internet with two routers connecting three LANs.*
- *Each device (computer or router) has a pair of addresses (**logical** and **physical**) for each connection.*
  - *In this case, **each computer** is connected to only one link and therefore has only one pair of addresses.*
  - ***Each router**, however, is connected to three networks (only two are shown in the figure). So each router has three pairs of addresses, one for each connection.*
- The computer with logical address A and physical address 10 needs to send a packet to the computer with logical address P and physical address 95.
- We use :
  - Letters (logical addresses)
  - Numbers → (physical addresses).

Figure 2.20 *IP addresses*



Example 2.3

- The sender encapsulates its data in a packet at the network layer and adds two logical addresses (A and P).
- The network layer consults its **routing table** and finds the logical address of the next hop (router 1) to be F
- The **ARP** finds the physical address of router 1 that corresponds to the logical address of F
- Now the network layer passes this address to the data link layer, which in turn, encapsulates the packet with physical destination address 20 and physical source address 10.

Example 2.3

- The frame is received by every device on LAN 1, but is discarded by all **except** router 1
- The router decapsulates the packet from the frame to read the logical destination address P
- Since the logical destination address does not match the router's logical address, the router knows that the packet needs **to be forwarded**.
- Router consults its **routing table** and **ARP** to find the physical destination address of the next hop (router 2)
- It creates a new frame, encapsulates the packet, and sends it to router 2

Example 2.3

- Note the physical addresses in the frame. The source physical address changes from 10 to 99. The destination physical address changes from 20 (router 1) to 33 (router 2).
- The **logical** source and destination addresses must **remain the same** (otherwise the packet will be lost.)
- At router 2 we have a similar scenario.
- The physical addresses are changed, and a new frame is sent to the destination computer.
- When the frame reaches the destination, the packet is decapsulated and delivered to the upper layer.

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The physical addresses will change from hop to hop,
but the logical addresses usually remain the same.

ADDRESSING

3. Port Addresses

- In the TCP/IP architecture, the label assigned to a **process** is called a port address.
- For example:
  - computer A can communicate with computer C by using **TELNET**. At the same time, computer A communicates with computer B by using the File Transfer Protocol (**FTP**). For these processes to receive data simultaneously, we need a method to label the different processes

Example 2.4

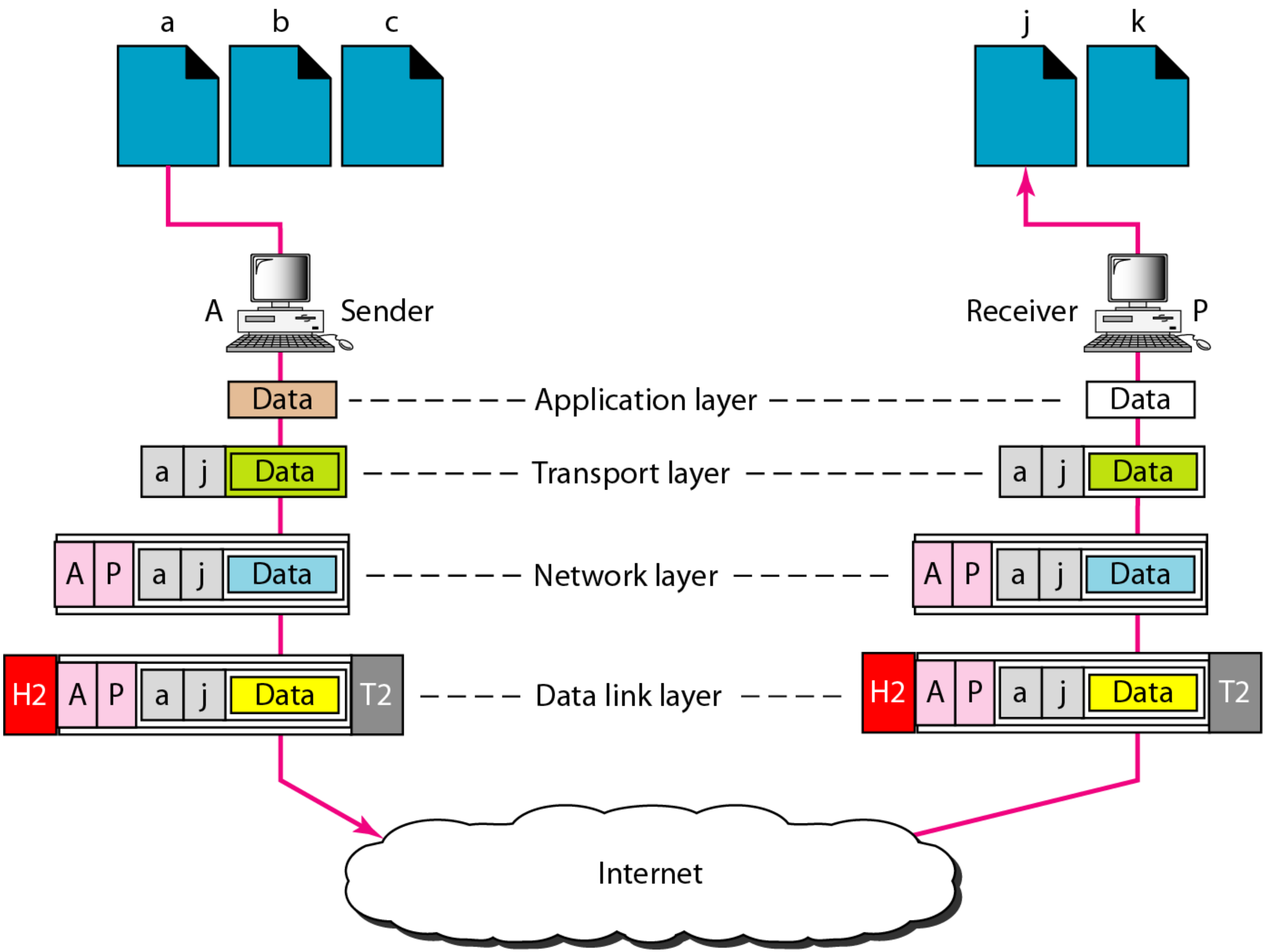
Figure 2.21 *shows two computers communicating via the Internet.*

The sending computer is running three processes at this time with port addresses a, b, and c.

The receiving computer is running two processes at this time with port addresses j and k.

Process a in the sending computer needs to communicate with process j in the receiving computer. Note that although physical addresses change from hop to hop, logical and port addresses remain the same from the source to destination.

Figure 2.21 *Port addresses*



Example 2.5

A port address is a 16-bit address represented by one ***decimal number*** *as shown.*

753

A 16-bit port address represented as one single number.

ADDRESSING

Specific Addresses

- Some applications have **user-friendly** addresses that are designed for that specific address.
- **Examples**:
  - e-mail address (for example, forouzan@fhda.edu)
  - Universal Resource Locator (URL) (for example, **www.mhhe.com**).
- These addresses, however, **get changed to** the corresponding **port** and **logical addresses** by the sending computer.